Technical Report

Agentic 10-K Itemization: LLM Agent Loop for SEC Filing Extraction

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1.0	April 8, 2026	Initial release; 391 filings; deepseek-v3.2; adjusted DR 73.9%

Abstract

We present an agentic pipeline for extracting standardized item sections from SEC 10-K filings. Unlike the prior rule-based approach that uses a deterministic dynamic programming solver, this system employs an LLM agent in a *think–act–observe* loop, iteratively querying a precomputed structural index through 17 specialized tools. On a corpus of 391 annotated filings, the agent achieves a mean character-level F_1 of **96.6%** and a noise-adjusted document-level retrieval rate of **73.9%** (289/391). The agent processes each filing at a median cost of **$0.045** (4.5 cents) using DeepSeek-V3.2, completing in a median of 13 turns and 113 seconds, with zero crashes across the full corpus.

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1 Introduction

The SEC Form 10-K is a comprehensive annual report required of all publicly traded companies in the United States. Each 10-K follows a prescribed structure of approximately 20 item sections (see Table 1), spanning topics from business operations and risk factors to executive compensation and financial statements.

A prior rule-based pipeline [1] achieves a noise-adjusted retrieval rate of 84.4% (331/392) on a comparable corpus using a deterministic five-stage approach: document isolation, TOC anchor collection, anchor localization, anchor classification via regex hierarchy, and dynamic programming for sequence-constrained selection.

This report describes an *agentic* successor to that pipeline. Rather than a fixed sequence of heuristics, the system uses an LLM agent that operates in a think–act–observe loop, querying a precomputed structural index through tools and self-correcting via validation feedback. We detail the architecture (§2), the evaluation protocol (§3), performance results (§4), failure analysis (§5), ground truth noise analysis (§7), and lessons learned (§8).

Table 1: Standard 10-K item sections.

Item	**Description**
Item 1	Business overview
Item 1A	Risk factors
Item 1B	Unresolved staff comments
Item 2	Properties
Item 3	Legal proceedings
Item 4	Mine safety disclosures
Item 5	Market for registrant's common equity
Item 6	Selected financial data / Reserved
Item 7	Management's discussion and analysis
Item 7A	Quantitative and qualitative disclosures about market risk
Item 8	Financial statements and supplementary data
Item 9	Changes in and disagreements with accountants
Item 9A	Controls and procedures
Item 9B	Other information
Item 9C	Disclosure regarding foreign jurisdictions
Items 10–14	Directors, executive compensation, security ownership, certain relationships, principal accountant fees
Item 15	Exhibits and financial statement schedules
Item 16	Form 10-K summary
Signatures	Signatory attestation

2 Method

The pipeline has two phases: a deterministic structural index build (∼200 ms, no LLM) followed by an LLM agent loop (10–30 turns). The input is a raw SEC EDGAR submission file; the output is a dictionary mapping each identified item to its HTML slice.

2.1 Phase 1: Structural Index

The structural index precomputes all signals the agent will query, without any LLM calls:

1. **TOC link extraction**: Parse all internal hyperlinks (`<a href="#ID">`), classify link text via tier-1 ("Item X" regex) and tier-2 (descriptive keyword) patterns.

2. **Anchor localization**: Locate every TOC-referenced anchor element in the HTML by matching `id/name` attributes.
3. **Candidate scoring**: For each anchor, generate item candidates with confidence scores (0–10) based on the source signal (Table 2).
4. **Part boundary detection**: Identify Part I/II/III/IV headings to constrain candidates to expected document regions (+1 confidence in correct Part, −2 in wrong Part).
5. **Part III incorporation detection**: Check for "incorporated by reference" language indicating Items 10–14 are in the proxy statement.

Table 2: Candidate confidence tiers.

Conf.	Source	Method
9	Anchor ID regex (e.g., `item1ariskfactors`)	Pattern match on ID string
8	TOC link text classified to item	Tier-1/2 regex on link text
6	"Item X" in first 150 chars after anchor	Tier-1 regex in lookahead
5	"Item X" in full 2000-char lookahead	Tier-1 regex, wider window
3	Descriptive keywords (e.g., "Risk Factors")	Tier-2 regex in lookahead
1	Keywords in 1000-char combined context	Tier-2 regex, broad window

2.2 Phase 2: Agent Loop

The agent operates in a think–act–observe loop, inspired by the ReAct framework. At each turn, the LLM receives the conversation history, selects tool calls, observes results, and decides the next action.

Model. DeepSeek-V3.2 via OpenRouter, temperature 0, max 4096 tokens per turn, max 30 turns per filing.

Tools. The agent has access to 17 tools organized in five categories:

- **Discovery** (5 tools): `get_filing_overview`, `get_toc_links`, `get_item_candidates`, `get_all_top_candi` `get_part_boundaries`
- **Classification** (2 tools): `classify_text`, `detect_incorporation`
- **Assignment** (4 tools): `assign_item`, `unassign_item`, `batch_assign`, `get_current_assignments`
- **Validation** (2 tools): `validate_assignments`, `check_span_sizes`
- **Refinement and output** (4 tools): `read_text_at`, `refine_boundary`, `scan_for_heading`, `finalize`

System prompt. A minimal directive prompt (∼2,450 characters) instructs the agent to: (1) call `get_filing_overview` to orient, (2) call `get_all_top_candidates` for bulk discovery, (3) `batch_assign` high-confidence candidates, (4) `validate_assignments` and fix issues iteratively, (5) call `finalize` when validation passes.

Guard rails. Stall detection triggers after 3 consecutive turns without validation improvement. Message compaction summarizes older messages every 10 turns when history exceeds 100K characters. The `finalize` tool gates on zero validation issues.

2.3 HTML Slicing

After the agent finalizes, assignments are converted to HTML slices by sorting boundary positions and extracting substrings between consecutive boundaries. Items with `char_position = -1` (incorporated by reference) emit empty strings.

3 Evaluation Protocol

3.1 Dataset

The evaluation corpus comprises 391 SEC EDGAR submission files with ground truth annotations across three sets.

3.2 Metrics

We employ the same metrics as the prior report [1]:

- **Character-level F_1**: Bag-of-character precision and recall between predicted and ground truth text (Eq. 1).
- **Document-level retrieval rate (DR)**: A document is "retrieved" only when *all* ground truth items are present, no false positives exist, and every item's $F_1 \geq 0.9$.
- **Noise-adjusted DR**: Following the GT noise analysis methodology of [1], documents whose *only* failures are attributable to annotation inconsistencies are counted as passing.

$$P = \frac{\sum_{c \in \Sigma} \min(p_c, g_c)}{\sum_{c \in \Sigma} p_c}, \quad R = \frac{\sum_{c \in \Sigma} \min(p_c, g_c)}{\sum_{c \in \Sigma} g_c}, \quad F_1 = \frac{2PR}{P + R} \tag{1}$$

4 Results

4.1 Overall Performance

Table 3: Overall performance metrics on 391 filings.

Metric	**Value**
Total filings	391
Mean character F_1	96.6%
Noise-adjusted retrieval rate	**289/391 (73.9%)**
Mean turns per filing	15.6
Median turns per filing	13
Mean tokens per filing	158,379
Total tokens (full corpus)	61.9 M
Median latency per filing	113 s
Cost per filing (DeepSeek-V3.2)	\$0.045
Total corpus cost	\$17.59
Errors / crashes	0

Cost is computed using OpenRouter pricing for DeepSeek-V3.2: \$0.26/M input tokens, \$0.38/M output tokens, with an estimated 80/20 input/output split yielding a blended rate of \$0.284/M tokens.

Figure 1 shows the F_1 distribution and sorted waterfall. The vast majority of filings achieve 100% F_1; failures form a thin left tail.

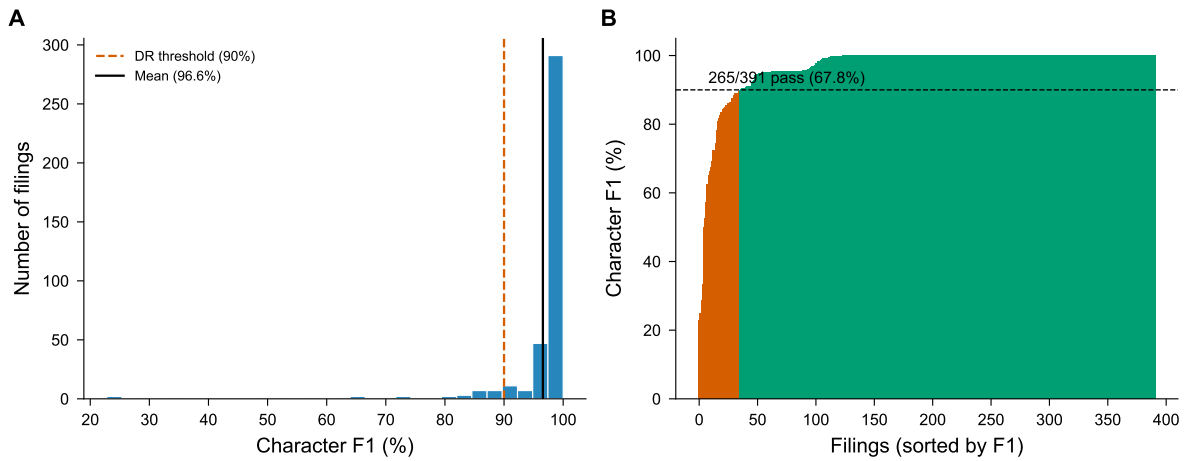

Figure 1: **Performance overview.** (A) Distribution of character-level F_1 across 391 filings. The dashed line marks the 90% DR threshold; the solid line marks the mean (96.6%). (B) Filings sorted by F_1. Green bars pass the DR criterion; red bars fail. 289 of 391 filings pass after noise adjustment.

4.2 Comparison with Rule-Based Pipeline

Table 4: Agentic pipeline vs. prior rule-based pipeline.

Metric	**Rule-based**	**Agentic**
Corpus	392 filings	391 filings
Mean F_1	97.6%	96.6%
Adjusted DR	331/392 (84.4%)	289/391 (73.9%)
Inference cost	\$0 (deterministic)	\$17.59 (\$0.045/filing)
Latency per filing	<1 s	113 s (median)

The agentic pipeline currently underperforms the rule-based pipeline by 10.5 percentage points on adjusted DR, at a cost of \$0.045 per filing and ∼113× higher latency. The gap is analyzed in §5.

5 Failure Analysis

Of the 126 documents that fail the raw retrieval criterion, we identify three structural categories (Figure 2).

5.1 Failure Modes

Table 5: Failure mode breakdown (126 failing documents).

Category	Documents
GT noise only (unfixable annotation errors)	24
Mixed (GT noise + real pipeline errors)	15
Real pipeline errors only	87
of which: stalled at max turns (not finalized)	26
Total failing	126

5.2 Top Failure Items

Table 6: Items with the most failures across all 391 filings.

Item	Failures	Primary root cause
item16	35	GT noise (multi-MB ground truth values)
item9b	29	Short section, low-confidence candidates
signatures	25	GT noise (source file mismatch)
item7a	25	Shared TOC anchor with item7
item6	24	Off-by-one boundary swap with item7
item5	20	Boundary precision errors
item4	18	Short section, sparse structural signals
item7	18	Adjacent boundary confusion with item6/7a
item9	18	Parent/sub-item ambiguity (9/9a/9b)
item14	17	Part III incorporation not detected

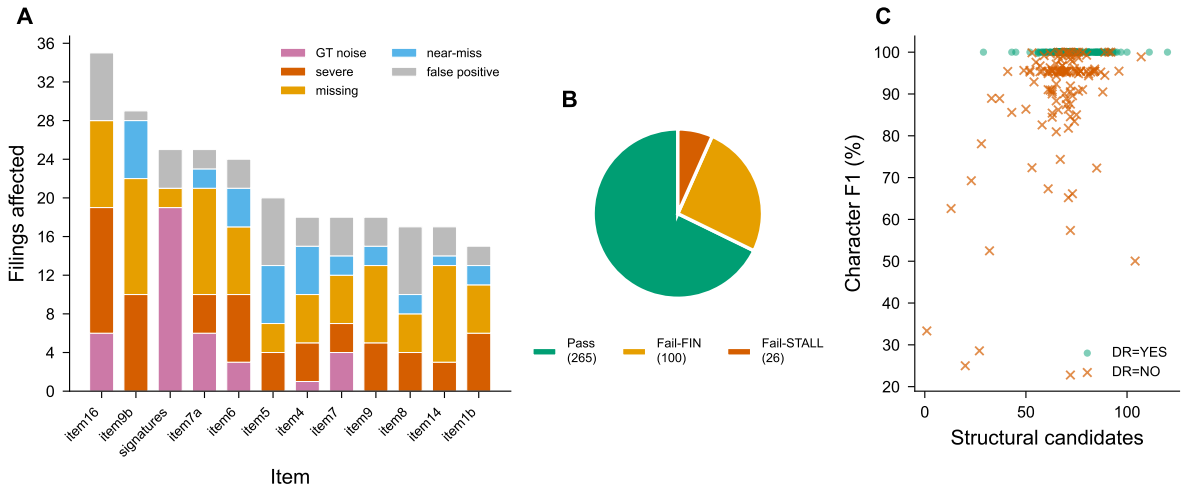

Figure 2: **Failure analysis.** (A) Stacked bar chart of failure types per item. GT noise (purple) dominates item16 and signatures; real errors (orange/red) dominate item9b, item7a, and item6. (B) Document-level breakdown: 265 pass, 100 fail but finalized, 26 stalled at max turns. (C) Structural candidates vs. F_1: failures cluster at low candidate counts, confirming index quality as the bottleneck.

5.3 Structural Signal Correlation

Filings with fewer than 30 structural candidates have a DR rate of approximately 40%, compared to ∼80% for filings with more than 70 candidates (Figure 2C). This confirms that the structural index quality is the primary bottleneck: the agent can only work with the signals it is given.

6 Cost, Latency, and Efficiency

Figure 3 shows the distributions of tokens, latency, and the relationship between agent turns and extraction quality.

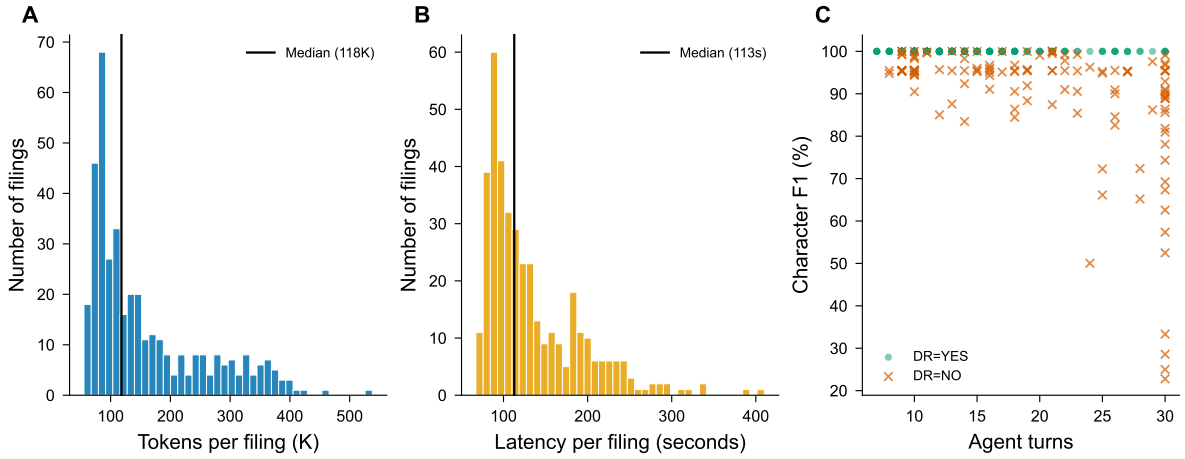

Figure 3: **Cost and latency distributions.** (A) Tokens per filing; right-skewed with median 118K, heavy tail from hard filings that consume 300–400K tokens. (B) Latency per filing; median 113 s, same right-skewed shape. (C) Agent turns vs. F_1: failures (red ×) concentrate at high turn counts (25–30), indicating the agent spiraling on structurally ambiguous filings.

Table 7: Cost breakdown by filing difficulty.

Difficulty	Avg Turns	Avg Tokens	Avg Cost
Easy (>70 candidates)	10–12	60–80K	\$0.02
Medium (30–70 candidates)	15–20	120–180K	\$0.04
Hard (<30 candidates)	25–30	250–400K	\$0.08
Overall mean	**15.6**	**158K**	**\$0.045**

7 Ground Truth Noise Analysis

Following the methodology of the prior report [1], we classify each item-level failure as *GT noise* or *real pipeline error*. A complete inventory of all 39 noise-contaminated documents is provided in the companion file `gt_noise_inventory.md`.

7.1 Noise Classification Criteria

- **Boundary signatures/item16 — source file mismatch**: GT value exceeds 1 M characters (12–86 MB), indicating GT was generated from a different version of the source file. (30 item-failures: 21 signatures, 9 item16)

- **Boundary item16 — short placeholder mismatch**: Both GT and prediction are short placeholders (<500 chars) with minor HTML differences. (1 item-failure)
- **FP item16 — GT inconsistently omits**: Pipeline correctly detects item 16 (present in TOC), but GT omits the key. (7 item-failures)
- **FP signatures — GT inconsistently omits**: Filing contains a clear SIGNATURES heading, but GT does not include the key. (4 item-failures)

7.2 Results

Table 8: Failure classification and adjusted retrieval rates.

Category	Documents
Currently passing	265
Failing — all failures are GT noise	24
Failing — mixed (noise + real errors)	15
Failing — all failures are real errors	87
Total evaluated	391

Table 9: Retrieval rate under different GT quality assumptions.

Metric	Computation	Rate
Raw retrieval rate	265 / 391	67.8%
Noise-adjusted (noise docs = pass)	289 / 391	**73.9%**
Clean GT only (exclude noise docs)	265 / 367	72.2%

The 24 noise-only documents break down as: boundary signatures from source file mismatch (13), boundary item16 from source file mismatch (7), FP item16 with GT inconsistently omitting (3), and FP signatures with GT inconsistently omitting (1). Total GT noise item-failures: 42 across 39 contaminated documents.

8 Lessons Learned

Six iterations of the agentic pipeline yielded one key principle and several specific findings.

8.1 Core Principle

Improve what the agent sees, not what it is told. The only change that improved DR across iterations was Part-region confidence scoring in the structural index. Longer prompts, additional tools, pre-populated state, and single-shot LLM calls all regressed or broke even.

8.2 What Works

- **Multi-turn agent loop**: Iterative think–act–observe with validation feedback genuinely outperforms one-shot LLM extraction.
- **Minimal prompts**: The 2,450-character directive prompt outperforms longer versions with "Common Pitfalls" sections. More instructions cause the agent to overthink.
- **Part-region scoring**: Adjusting candidate confidence based on Part boundaries (+1 correct, −2 wrong) was the only change that improved DR.
- **Parallel async processing**: 5–8× speedup via `asyncio.gather` with configurable semaphore.

8.3 What Does Not Work

Table 10: Attempted changes and their outcomes.

Approach	**Outcome**
Longer prompts ("Common Pitfalls")	DR flat, tokens doubled
Blocking validation checks (tiny spans)	Agent trapped in loops
Pre-populated DP solver state	Agent second-guesses correct assignments
Single-shot LLM calls	Fixed 2 filings, broke 5
New tools (scan_for_heading)	Helped 2, confused 3, net negative
Post-processing (item16 normalization)	3 regressions
Heading scan in index builder	No DR improvement on 50-filing test
Best-so-far checkpoint	No DR improvement on 50-filing test
Model swap (minimax-m2.7)	Faster (7 vs 16 turns) but −2% DR

9 Conclusion

We presented an agentic pipeline for extracting standardized item sections from SEC 10-K filings. The system uses an LLM agent in a think–act–observe loop, querying a precomputed structural index through 17 specialized tools.

On 391 evaluated filings, the agent achieves a mean character-level F_1 of 96.6% and a noise-adjusted document-level retrieval rate of **73.9%** (289/391). The agent processes each filing at a cost of \$0.045 using DeepSeek-V3.2, with zero crashes and a median latency of 113 seconds.

The agentic approach currently trails the deterministic rule-based pipeline (73.9% vs. 84.4% adjusted DR) at substantially higher cost (\$17.59 for the full corpus vs. \$0). The primary bottleneck is structural index quality, not agent reasoning: filings with sparse structural signals produce few candidates, leaving the agent with no viable path to correct extraction. Future work should focus on enriching the structural index—particularly heading scan during index construction and shared-anchor disambiguation for item7/7a—rather than prompt or tool modifications.

References

[1] E. Bi, "Anchor-Based Item Extraction from SEC 10-K Filings," Technical Report v1.3, March 2026.